

# Pratham Raj

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[pratham2411.online](https://pratham2411.online) | [LeetCode: prathamraj2411](https://leetcode.com/prathamraj2411) | [Codeforces: prathamraj2411](https://codeforces.com/prathamraj2411)

## EDUCATION

### National Institute of Technology Patna

B.Tech in Electrical Engineering

Patna, India

Aug 2023 – May 2027

## TECHNICAL SKILLS

**Languages:** C/C++, Python, Java, JavaScript (ES6+), TypeScript, HTML5, CSS3  
**Backend & APIs:** Node.js, Express.js, RESTful APIs, cpp-http lib, JWT Authentication, Firebase Authentication  
**Frontend:** React.js, Zustand, Tailwind CSS, Responsive Design, PWA, @dnd-kit  
**Databases & Cloud:** MongoDB, Cloudinary, Vercel, Render, Mongoose ODM, Firebase  
**Tools & Practices:** Git/GitHub, Linux/Unix, CMake, Agile, Code Review  
**Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Multi-threading, Scalability

## EXPERIENCE

### Software Engineering Contributor

Snorkel AI (via HireArt)

Jul 2026 – Present

Remote

- Reviewed and validated software engineering tasks sourced from real-world **GitHub pull requests**, evaluating each for solvability, clarity, and technical correctness across Python and JavaScript codebases.
- Authored and refined **outcome-based test suites** (10+ fail-to-pass tests per task) to verify code behavior end-to-end, close reward-hacking loopholes, and enforce strict test-to-requirement traceability.
- Collaborated with a distributed team of engineering reviewers through an **adjudication workflow** to resolve submission discrepancies, maintaining rigorous quality standards for AI coding-agent training datasets.

## PROJECTS

### NIT Patna Market | *React.js, Node.js, MongoDB, Cloudinary, JWT* | [GitHub](#) | [Live](#)

Aug 2025 – Present

- Led full-stack **MERN** development with **JWT authentication, role-based access control** (guest/user/seller/admin), and OTP-verified `@nitp.ac.in` signup via Resend API, securely serving **1,000+ university students**.
- Built dual-mode **image pipeline** (Cloudinary CDN + local-disk fallback) for up to 8 images/listing, reducing image load times by **30%**; buyer-seller **HTTP polling chat** with read receipts and cron-scheduled email digest.
- Deployed as a **PWA** on Vercel + Render with optimistic UI enabling **500+ daily transactions**; indexed **MongoDB** queries with Mongoose ODM achieving sub-100ms response times.

### VectorForge – C++ Vector Database & RAG Engine | *C++17, HNSW, KD-Tree, cpp-http lib, Ollama* | [GitHub](#)

Jan 2026 – Present

- Built a **vector database from scratch** in C++ implementing three interchangeable K-NN algorithms – Brute Force  $O(N)$ , KD-Tree, and **HNSW** ( $M=16$ ,  $ef_{construction}=200$ ) with Cosine, Euclidean, and Manhattan distance metrics, demonstrating how Pinecone, Weaviate, and Chroma work under the hood.
- Implemented a **RAG pipeline**: text → Ollama (nomic-embed-text, **768D embeddings**) → HNSW semantic search → **Llama 3.2** generation, with 250-word overlapping chunking across multi-document stores.
- Exposed a **REST API** via `cpp-http lib` (insert, delete, search, benchmark, `hnsw-info` endpoints) with a live **2D PCA scatter plot** visualizing semantic clusters of 20 pre-loaded vectors across 4 categories.

## ACHIEVEMENTS & COMPETITIVE PROGRAMMING

**Amazon HackOn 6.0:** Ranked **Top 75 coders** globally out of **75,000+** participants.

**Competitive Programming: Knight Badge** (2000+ peak rating) on LeetCode | **3-Star** (1650+) on CodeChef | **Pupil** (1250+) on Codeforces | **2000+** problems solved.

**Hackathon Champion:** Secured **1st place** at Hackslash College Hackathon – rapid prototyping and algorithmic optimization under strict time constraints.

**Industry Recognition:** Shortlisted for **Digital Specialist Engineer (DSE)** at Infosys via top-percentile technical assessments. Ranked **Top 10** institution-wide on Codolio leaderboard at NIT Patna.

## LEADERSHIP & VOLUNTEERING

**Campus Ambassador, HackerRank & GfG:** Facilitated DSA workshops, mentoring peers in competitive programming and algorithmic thinking.

**Educational Volunteer, NSS:** Instructed **500+** students, spearheading community-driven educational outreach initiatives.

**Technical Contributor, ELUSOC:** Architected open-source software solutions, collaborating to foster campus developer communities.